# INTRODUCTION TO MACHINE LEARNING CONCEPTS

## (30 MINUTES)



Machine learning is the basis for most modern artificial intelligence solutions. A familiarity with the core concepts on which machine learning is based is an important foundation for understanding AI.

**Learning objectives**

After completing this module, you will be able to:

- Describe core concepts of machine learning
- Identify different types of machine learning
- Describe considerations for training and evaluating machine learning models
- Describe core concepts of deep learning

### 1.0 Introduction

Machine learning is in many ways the intersection of two disciplines - data science and software engineering. The goal of machine learning is to use data to create a predictive model that can be incorporated into a software application or service. To achieve this goal requires collaboration between data scientists who explore and prepare the data before using it to train a machine learning model, and software developers who integrate the models into applications where they're used to predict new data values (a process known as inferencing).

Machine learning has its origins in statistics and mathematical modeling of data. The fundamental idea of machine learning is to use data from past observations to predict unknown outcomes or values. For example:

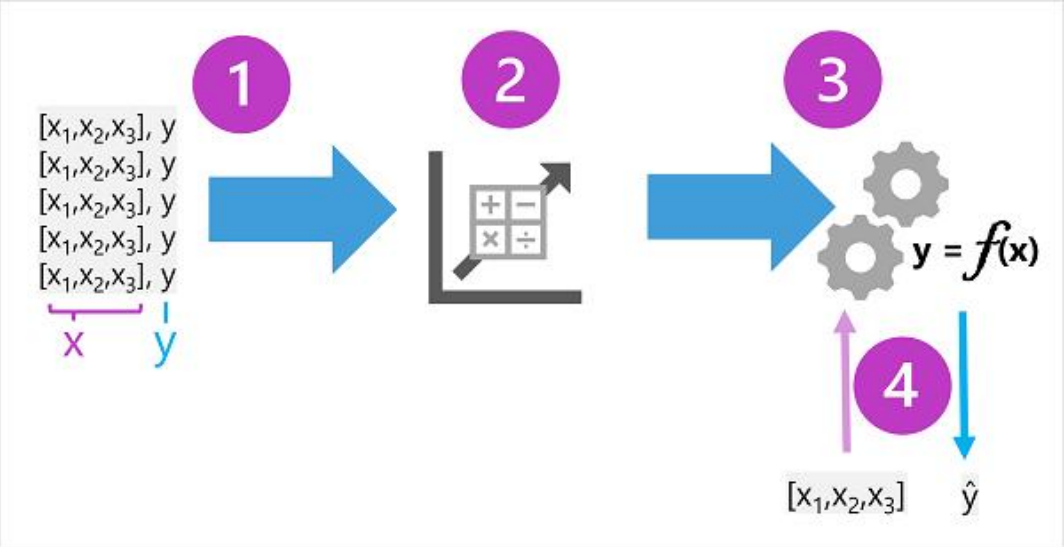
- The proprietor of an ice cream store might use an app that combines historical sales and weather records to predict how many ice creams they're likely to sell on a given day, based on the weather forecast.
- A doctor might use clinical data from past patients to run automated tests that predict whether a new patient is at risk from diabetes based on factors like weight, blood glucose level, and other measurements.
- A researcher in the Antarctic might use past observations to automate the identification of different penguin species (such as Adelie, Gentoo, or Chinstrap) based on measurements of a bird's flippers, bill, and other physical attributes.

### 2.0 Machine Learning Models

Because machine learning is based on mathematics and statistics, it's common to think about machine learning models in mathematical terms. Fundamentally, a

machine learning model is a software application that encapsulates a function to calculate an output value based on one or more input values. The process of defining that function is known as training. After the function has been defined, you can use it to predict new values in a process called inference.

Let's explore the steps involved in training and inference.




1. The training data consists of past observations. In most cases, the observations include the observed attributes or features of the thing being observed, and the known value of the thing you want to train a model to predict (known as the label).

   In mathematical terms, you'll often see the features referred to using the shorthand variable name **x**, and the label referred to as **y**. Usually, an observation consists of multiple feature values, so **x** is actually a vector (an array with multiple values), like this: **[x1,x2,x3,...]**.

   To make this clearer, let's consider the examples described previously:

   - In the ice cream sales scenario, our goal is to train a model that can predict the number of ice cream sales based on the weather. The weather measurements for the day (temperature, rainfall, wind-speed, and so on) would be the features (**x**), and the number of ice creams sold on each day would be the label (**y**).
   - In the medical scenario, the goal is to predict whether or not a patient is at risk of diabetes based on their clinical measurements. The patient's measurements (weight, blood glucose level, and so on) are the features (**x**), and the likelihood of diabetes (for example, 1 for at risk, 0 for not at risk) is the label (**y**).
   - In the Antarctic research scenario, we want to predict the species of a penguin based on its physical attributes. The key measurements of the

penguin (length of its flippers, width of its bill, and so on) are the features (**x**), and the species (for example, 0 for Adelie, 1 for Gentoo, or 2 for Chinstrap) is the label (**y**).

2. An algorithm is applied to the data to try to determine a relationship between the features and the label, and generalize that relationship as a calculation that can be performed on **x** to calculate **y**. The specific algorithm used depends on the kind of predictive problem you're trying to solve (more about this later), but the basic principle is to try to fit the data to a function in which the values of the features can be used to calculate the label.

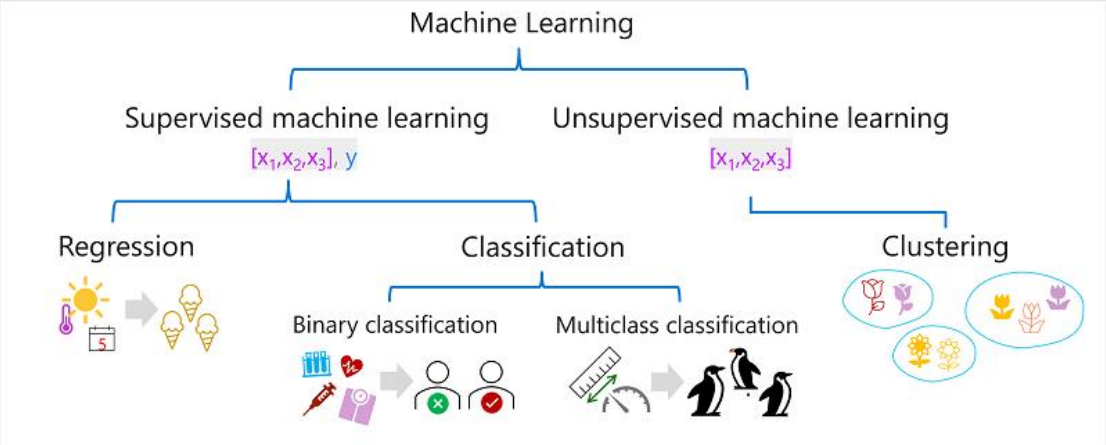
3. The result of the algorithm is a model that encapsulates the calculation derived by the algorithm as a function - let's call it **f**. In mathematical notation:

   **y = f(x)**

4. Now that the training phase is complete, the trained model can be used for inferencing. The model is essentially a software program that encapsulates the function produced by the training process. You can input a set of feature values, and receive as an output a prediction of the corresponding label. Because the output from the model is a prediction that was calculated by the function, and not an observed value, you'll often see the output from the function shown as **ŷ** (which is rather delightfully verbalized as "y-hat").

### 3.0 Types of Machine Learning Models

There are multiple types of machine learning, and you must apply the appropriate type depending on what you're trying to predict. A breakdown of common types of machine learning is shown in the following diagram.




**Supervised machine learning**

*Supervised* machine learning is a general term for machine learning algorithms in which the training data includes both *feature* values and known *label* values.

Supervised machine learning is used to train models by determining a relationship between the features and labels in past observations, so that unknown labels can be predicted for features in future cases.

## Regression

Regression is a form of supervised machine learning in which the label predicted by the model is a numeric value. For example:

- The number of ice creams sold on a given day, based on the temperature, rainfall, and wind-speed.
- The selling price of a property based on its size in square feet, the number of bedrooms it contains, and socioeconomic metrics for its location.
- The fuel efficiency (in miles-per-gallon) of a car based on its engine size, weight, width, height, and length.

## Classification

Classification is a form of supervised machine learning in which the label represents a categorization, or *class*. There are two common classification scenarios.

### *Binary classification*

In binary classification, the label determines whether the observed item is (or isn't) an instance of a specific class. Or put another way, binary classification models predict one of two mutually exclusive outcomes. For example:

- Whether a patient is at risk for diabetes based on clinical metrics like weight, age, blood glucose level, and so on.
- Whether a bank customer will default on a loan based on income, credit history, age, and other factors.
- Whether a mailing list customer will respond positively to a marketing offer based on demographic attributes and past purchases.

In all of these examples, the model predicts a binary true/false or positive/negative prediction for a single possible class.

### *Multi-class classification*

Multi-class classification extends binary classification to predict a label that represents one of multiple possible classes. For example,

- The species of a penguin (Adelie, Gentoo, or Chinstrap) based on its physical measurements.
- The genre of a movie (comedy, horror, romance, adventure, or science fiction) based on its cast, director, and budget.

In most scenarios that involve a known set of multiple classes, multi-class classification is used to predict mutually exclusive labels. For example, a penguin can not be both a Gentoo and an Adelie. However, there are also some algorithms that you can use to train multi-label classification models, in which there may be more than one valid label for a single observation. For example, a movie could potentially be categorized as both science fiction and comedy.

## Unsupervised machine learning

*Unsupervised* machine learning involves training models using data that consists only of *feature* values without any known labels. Unsupervised machine learning algorithms determine relationships between the features of the observations in the training data.

### Clustering

The most common form of unsupervised machine learning is clustering. A clustering algorithm identifies similarities between observations based on their features, and groups them into discrete clusters. For example:

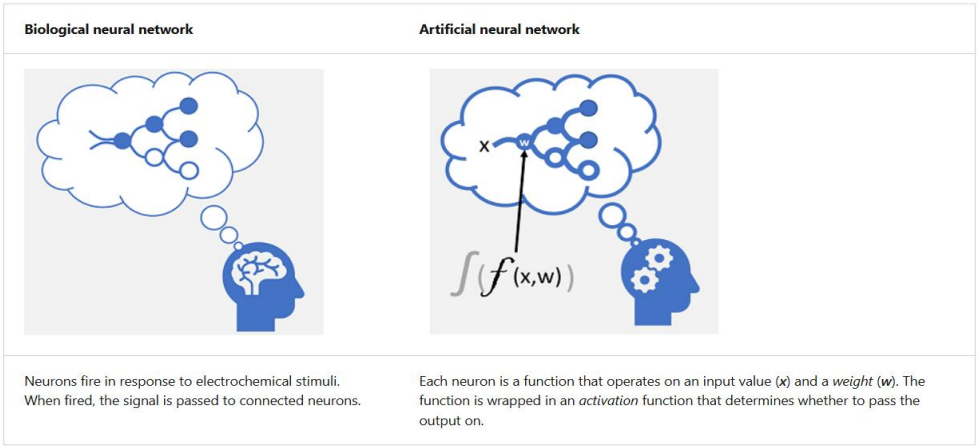
- Group similar flowers based on their size, number of leaves, and number of petals.
- Identify groups of similar customers based on demographic attributes and purchasing behavior.

In some ways, clustering is similar to multi-class classification; in that it categorizes observations into discrete groups. The difference is that when using classification, you already know the classes to which the observations in the training data belong; so the algorithm works by determining the relationship between the features and the known classification label. In clustering, there's no previously known cluster label and the algorithm groups the data observations based purely on similarity of features.

In some cases, clustering is used to determine the set of classes that exist before training a classification model. For example, you might use clustering to segment your customers into groups, and then analyze those groups to identify and categorize different classes of customer (high value - low volume, frequent small purchaser, and so on). You could then use your categorizations to label the observations in your clustering results and use the labeled data to train a classification model that predicts to which customer category a new customer might belong.

## Deep learning

*Deep learning* is an advanced form of machine learning that tries to emulate the way the human brain learns. The key to deep learning is the creation of an artificial *neural network* that simulates electro-chemical activity in biological neurons by using mathematical functions, as shown here.



Artificial neural networks are made up of multiple layers of neurons - essentially defining a deeply nested function. This architecture is the reason the technique is referred to as deep learning and the models produced by it are often referred to as deep neural networks (DNNs). You can use deep neural networks for many kinds of machine learning problem, including regression and classification, as well as more specialized models for natural language processing and computer vision.

Just like other machine learning techniques discussed in this module, deep learning involves fitting training data to a function that can predict a label (**y**) based on the value of one or more features (**x**). The function (**f(x)**) is the outer layer of a nested function in which each layer of the neural network encapsulates functions that operate on **x** and the weight (**w**) values associated with them. The algorithm used to train the model involves iteratively feeding the feature values (**x**) in the training data forward through the layers to calculate output values for $\hat{\mathbf{y}}$, validating the model to evaluate how far off the calculated $\hat{\mathbf{y}}$ values are from the known **y** values (which quantifies the level of error, or loss, in the model), and then modifying the weights (**w**) to reduce the loss. The trained model includes the final weight values that result in the most accurate predictions.